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Polycystic ovary syndrome

Elisabet Stener-Victorin¹✉, Helena Teede², Robert J. Norman³, Richard Legro^{4,5}, Mark O. Goodarzi⁶, Anuja Dokras⁷, Joop Laven⁸, Kathleen Hoeger⁹ & Terhi T. Piltonen¹⁰

Abstract

Despite affecting ~11–13% of women globally, polycystic ovary syndrome (PCOS) is a substantially understudied condition. PCOS, possibly extending to men's health, imposes a considerable health and economic burden worldwide. Diagnosis in adults follows the International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome, requiring two out of three criteria – clinical or biochemical hyperandrogenism, ovulatory dysfunction, and/or specific ovarian morphological characteristics or elevated anti-Müllerian hormone. However, diagnosing adolescents omits ovarian morphology and anti-Müllerian hormone considerations. PCOS, marked by insulin resistance and hyperandrogenism, strongly contributes to early-onset type 2 diabetes, with increased odds for cardiovascular diseases. Reproduction-related implications include irregular menstrual cycles, anovulatory infertility, heightened risks of pregnancy complications and endometrial cancer. Beyond physiological manifestations, PCOS is associated with anxiety, depression, eating disorders, psychosexual dysfunction and negative body image, collectively contributing to diminished health-related quality of life in patients. Despite its high prevalence persisting into menopause, diagnosing PCOS often involves extended timelines and multiple health-care visits. Treatment remains ad hoc owing to limited understanding of underlying mechanisms, highlighting the need for research delineating the aetiology and pathophysiology of the syndrome. Identifying factors contributing to PCOS will pave the way for personalized medicine approaches. Additionally, exploring novel biomarkers, refining diagnostic criteria and advancing treatment modalities will be crucial in enhancing the precision and efficacy of interventions that will positively impact the lives of patients.

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A full list of affiliations appears at the end of the paper. ✉ e-mail: elisabet.stener-victorin@ki.se

Introduction

Polycystic ovary syndrome (PCOS) is a complex hormonal condition characterized by signs of elevated androgens, irregular menstruation and polycystic ovaries. PCOS is an understudied yet major health threat for women and, possibly, for men (Box 1). Worldwide, ~11–13% of women suffer from PCOS¹, resulting in a substantial health and economic burden globally^{2–4}.

Based on the International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome¹, PCOS diagnosis in adults requires at least two of the following three criteria – clinical hyperandrogenism and/or biochemical hyperandrogenism with elevated total testosterone, free androgen index, or bioavailable testosterone; ovulatory dysfunction with irregular or absent menstrual cycles; or polycystic ovaries with multiple small antral follicles and/or enlarged ovaries and/or elevated anti-Müllerian hormone (AMH) levels¹. In adolescents, a definitive diagnosis requires hyperandrogenism and irregular cycles but does not include ovarian morphology or AMH levels as they are a normal part of development in adolescents.

PCOS, hyperandrogenism and insulin resistance are strong contributors to the development of early-onset type 2 diabetes mellitus (T2DM)^{5,6}, with preceding hyperinsulinaemia and insulin resistance⁷. The prevalence of cardiometabolic risk factors, including excess weight, impaired glucose tolerance, T2DM, dyslipidaemia, hypertension and metabolic syndrome^{8–11}, are consistently higher in women with PCOS, likely contributing to the increased odds for cardiovascular disease, including ischaemic heart disease, myocardial infarction, stroke and, potentially, cardiovascular mortality^{12–14}.

In individuals of reproductive age, PCOS is the most common cause of irregular menstrual cycles and anovulatory infertility¹. Moreover, PCOS causes considerably increased but poorly appreciated pregnancy complications, including early miscarriage¹, gestational diabetes, hypertension and pre-eclampsia, and is associated with a markedly increased risk of endometrial cancer^{15–17}.

Despite the high prevalence of PCOS, with hyperandrogenism and hyperinsulinaemia persisting even after menopause¹³, it can take

years and several different doctor visits before diagnosis¹⁸. Moreover, the treatment of PCOS is often ad hoc and is hampered by the lack of knowledge regarding the aetiology and underlying mechanisms of the syndrome. We do know that hyperandrogenaemia and hyperinsulinaemia have key pathogenetic roles and that PCOS runs in families, with an estimated heritability of 70%^{6,19,20}. However, how PCOS is inherited remains unclear; increasing evidence suggests that genetic and epigenetic mechanisms act in concert to lead to a complex trait such as PCOS, similar to T2DM²¹.

The prevalence of anxiety and depression is increased in women with PCOS, with increased eating disorders, psychosexual dysfunction and negative body image likely contributing to the severity of depression and anxiety. Overall, the clinical features, together with these associated complications, contribute to poor health-related quality of life (HRQoL) in patients^{22–24}.

In this Primer, we highlight the latest advances related to diagnosis, prevalence, associated comorbidities and current management of PCOS. In addition, we identify knowledge gaps in the underlying aetiology, pathophysiology and associated comorbidities of PCOS to guide future research to improve diagnosis and screening and to develop preventive and novel therapeutic strategies.

Epidemiology

Prevalence

Being the most common lifelong endocrinopathy in women, the criteria to diagnose PCOS has evolved from being a consensus Rotterdam criteria²⁵ to an evidence-based guideline criteria¹. However, controversy surrounds the inclusion of polycystic ovarian morphology (PCOM) in diagnostic criteria and relates to the lack of specificity of PCOM, the continuum of follicle counts without a clear delineating cut point, changes with age, methodological limitations with evolving technologies, and challenges with reporting²⁶.

Based on the Rotterdam criteria, population-based studies indicate a prevalence of 14–19%^{27,28}, although studies from the 2000s provided prevalence estimates of 10–13%²⁴. Moreover, individuals with

Box 1

Male phenotype of PCOS

Although studies implicate that a male equivalent of polycystic ovary syndrome (PCOS) exists, especially among men with a mother and/or sister with the syndrome, no diagnostic criteria are available. Furthermore, the mechanisms causing observed endocrine and metabolic alterations in men are unclear²⁹¹. The early onset of androgenic alopecia (before 35 years of age) has been observed in men with a family history of PCOS²⁹². Moreover, men with early androgenic alopecia exhibit elevated circulating testosterone, dehydroepiandrosterone sulfate, and luteinizing hormone and decreased follicle-stimulating hormone and sex hormone-binding globulin levels²⁹³. Reduced levels of follicle-stimulating hormone and sex hormone-binding globulin may be potentially linked to insulin resistance²⁹⁴ — features equivalent to those in women with PCOS.

Consistent with these findings, sons of women with PCOS have greater obesity than sons of women without PCOS with lipid

dysfunction, suggestive of latent insulin resistance²⁹⁵. Moreover, brothers of women with PCOS have increased anti-Müllerian hormone levels, altered gonadotrophin and steroidogenic secretion^{296,297}, and a metabolic phenotype characterized by dyslipidaemia, insulin resistance and pancreatic β -cell dysfunction as well as an increased cardiovascular disease risk^{298–301}. Current evidence indicates that genetic risk factors for PCOS contribute to an increased likelihood of obesity, type 2 diabetes mellitus, cardiovascular disease and androgenic alopecia in men⁹⁶, aligning with earlier observations of familial clustering of the metabolic phenotype in brothers of women with PCOS³⁰².

Although genetic components likely play a role in the male PCOS phenotype, clinical observations suggest that maternal obesity and PCOS may also impact the development of male fetuses, predisposing them to reproductive and metabolic disorders later in life, as observed in their female siblings^{106,109}.

Life course in women with PCOS

Adolescence	Reproductive age	Premenopause	Postmenopause
- PCOS symptoms - Increasing BMI - Psychological burden	- PCOS symptoms - Subfertility - Pregnancy complications - Adverse pregnancy-related outcomes - Higher BMI and metabolic burden - Psychological burden - Other health conditions start accumulating		- PCOS symptoms mitigate but still present - Higher BMI and metabolic burden - Psychological burden - Other health conditions accumulate

Fig. 1 | Life course in women with PCOS. Illustration of life course trajectory of polycystic ovary syndrome (PCOS), from adolescence to postmenopause.

PCOS exhibit geographic and ethnic variability in symptom presentation, for example, Hispanic and African American people with PCOS have greater insulin resistance than other ethnicities^{29,30}. AMH levels have been shown to correlate with PCOM and are now recommended in international guidelines for adult PCOS diagnosis with moderate evidence certainty^{26,28}. However, AMH levels have limitations similar to PCOM, with a lack of clear cut-off points for PCOS, changes with age, variable assays, and challenges in reporting owing to a lack of global standards and limited evidence across those of different ancestry.

Obesity exacerbates endocrine abnormalities and symptoms in individuals with PCOS^{30,31} and increases prevalence³². Moreover, one Chinese study suggests that the prevalence of PCOS increases with age, being ~10% in adolescents between 10 and 20 years of age and 17% in women 21–30 years of age²⁹. Two population-based studies have also reported such high prevalence in adulthood^{27,28}. Owing to the decline in ovarian reserve and androgen levels towards menopause, the number of women presenting with PCOS symptoms or findings diminishes, thereby lowering the prevalence of the syndrome with increasing age³³. However, given that irregular menstrual cycles and PCOM are physiological in adolescents, evidence and guidelines recommend not including these features in adolescents to avoid overdiagnosis. Although the underlying syndrome persists, ageing women with PCOS may eventually get increasingly regular cycles and PCOM is lost, making diagnosis challenging in older women (>50 years of age)²⁹. However, long-term data suggests that PCOS-related comorbidity persists to perimenopausal years and beyond^{34,35}. Hence, PCOS prevalence in fertile age likely reflects the true overall prevalence (Fig. 1).

Comorbidities

Obesity. Obesity is a common comorbidity in PCOS, affecting 20–80% of patients, where clinical cohorts tend to capture those with higher weight than non-selected populations^{26,31,36}. In population-based studies, clinical measures of obesity include BMI, waist circumference and waist-to-hip ratio³⁷. Many of these indicators are elevated in PCOS, although no optimal indicator has been established³⁸. The development of overweight in PCOS starts early, possibly before 5 years of age, supporting early onset of the syndrome³⁹. Weight gain is particularly pronounced from puberty to adulthood, for example, 11% of individuals with PCOS have exhibited a high BMI trajectory, compared with 6.6% in controls without PCOS, resulting in a +2.6 kg difference over a 10-year follow-up^{31,40}. However, the weight gap narrows towards menopause, as women without PCOS also gain weight. Ethnic differences exist in terms of weight gain associated with PCOS. For instance, American Hispanic people, African Americans and South Pacific Islanders show the highest weight, whereas Europeans and South or East Asians have low BMIs but more central obesity⁴¹. Of note, different BMI cut-offs

for overweight and obesity apply to Asian populations, warranting consideration when making cross-ethnic comparisons⁴².

Hyperandrogenism. Epidemiological data indicates that 40–80% of women with PCOS exhibit hyperandrogenism²⁴. The clinical manifestations of hyperandrogenism vary depending on ethnicity, obesity and age. Hirsutism (excess hair growth) affects ~70–80% of patients compared with 13% (range 8–20%) in the general population, with geographic variations²⁴. Acne prevalence is ~43% (95% CI 41–45%) in individuals with PCOS compared with 21% (95% CI 19–22%) in those without PCOS, with a higher rate of acne in adolescents than in adults or those with hyperandrogenic PCOS, with some geographic variation⁴³. Female pattern hair loss prevalence is estimated at 28% (95% CI 22–34%) in women with PCOS⁴⁴. Importantly, these manifestations are associated with substantial psychological distress, emphasizing the need for comprehensive treatment of women with the syndrome⁴⁵.

Metabolic dysfunction. Approximately 50–80% of women with PCOS exhibit insulin resistance^{46,47}. These individuals face an elevated risk of pre-diabetes and up to fourfold increased risk for T2DM, with the condition developing at an earlier age than in women without PCOS³⁵. Although T2DM risk is increased in individuals with PCOS regardless of BMI, variations in risk for insulin resistance and T2DM are noted in individuals with different BMI, geographical locations and ethnicities^{41,48}.

Cardiovascular disease. Current analyses have strengthened the evidence for increased risk for different cardiovascular disease outcomes in individuals with PCOS^{13,49}. Large data sets now indicate potentially increased risk for major cardiovascular end points^{12,35}, including myocardial infarction, stroke, heart failure and cardiovascular mortality^{12,35,50}. The analysis showed the hazard ratio (HR) for the composite end point (that is, when combining two or more end points of interest within a single variable) of 1.26 (95% CI 1.13–1.41) in a UK data set and 2.33 (1.26–4.30, Rotterdam criteria) and 2.47 (1.18–5.17, NIH criteria) in population-based data from Finland³⁵. Importantly, this risk seems to accumulate at a relatively young age (~30–35 years onwards), even before menopause. Although the absolute risk is low as the populations assessed have been relatively young (in most studies, women were aged <55 years), evidence from systematic reviews resulted in updated guideline recommendations, which call for regular screening and early prevention as well as research conducted in women in postmenopause¹.

Metabolic dysfunction-associated steatotic liver disease. Studies have demonstrated a close link between PCOS and major risk factors for the development of liver disease, including obesity, dyslipidaemia, insulin resistance, cardiovascular disease and hyperandrogenism⁵¹.

One meta-analysis indicates a 43% increased prevalence of metabolic dysfunction-associated steatotic liver disease (MASLD)⁵² (95% CI 35–52%) in women with PCOS⁵³. Further analyses show that the increased MASLD prevalence in these women could potentially be mediated by a high prevalence of metabolic syndrome, elevated homeostatic model assessment of insulin resistance (HOMA-IR), free androgen index and circulating testosterone⁵³. Although MASLD is highly prevalent in relatively young women with PCOS, absolute risk is still low and evidence is yet not strong enough to justify screening. Thus, further research into underlying aetiology and pathophysiology is needed.

Infertility. Infertility risk is 15 times higher in women with PCOS than in women without PCOS^{54,55}, yet the likelihood of having at least one child is similar between both groups of women⁵⁶. Counselling should emphasize the potential for pregnancy, though it may take time, often requiring fertility treatments, especially in individuals with excess weight or obesity^{54,55,57}. Spontaneous ovulation can occur, necessitating contraception if pregnancy is not desired. As women with PCOS reach ~35–40 years of age, regular cycles and possible ovulation can emerge, but the implications for later fertility remain uncertain.

Obstetric complications. Women with PCOS face a twofold to threefold increased risk of pregnancy complications, including miscarriage, pregnancy-induced hypertension, pre-eclampsia, gestational diabetes, prematurity and small-for-gestational-age offspring⁵⁸. These risks underscore the long-term health implications in daughters and likely also sons born to women with the syndrome. Although gestational diabetes risk is independent of BMI, a higher BMI increases the risk in certain white populations^{59,60}. By contrast, pregnancy-induced hypertension and pre-eclampsia seem to have stronger, BMI-independent effects⁵⁸.

Psychological disorders. PCOS is associated with a high prevalence of mental health disorders. Independent of obesity, diagnostic criteria or geographical region, 36.6% of women with PCOS present with moderate-to-severe depressive symptoms and 41.8% present with anxiety symptoms compared with 13.2% and 8.5%, respectively, in controls⁸. Adolescents also have higher odds of depression (OR 2.26, 95% CI 1.36–3.76) than controls, although evidence is limited for anxiety symptoms according to the updated evidence-based guideline²⁴. Nevertheless, a systematic review and meta-analysis demonstrated that symptoms of both anxiety and depression are more prevalent amongst adolescents with PCOS than amongst adolescents without PCOS²³. Moreover, the perinatal and postpartum periods may present an additional risk for depression⁶¹. Individuals with PCOS also have increased odds of binge eating disorder (OR 1.53, 95% CI 1.29–1.82)²⁴. PCOS-specific features that impact feminine identities, such as excess weight, hirsutism and acne, lead to increased body image distress, reduced sexual function and decreased sexual satisfaction^{24,62}. Importantly, body image distress may be a mediator of anxiety and depressive symptoms in individuals with PCOS⁶¹.

Cancer. PCOS is associated with a twofold overall increased risk for cancer⁶³, with the strongest evidence supporting elevated endometrial cancer risk (fivefold) owing to anovulation, progesterone resistance, hyperinsulinaemia, overweight and low-grade inflammation^{15,64,65}. Although Mendelian randomization analyses have linked PCOS to breast cancer^{66,67}, clinical evidence is lacking^{13,68}. Possible protective effects of anovulation on breast tissue have not been studied. Findings

from some studies suggest an increased risk of ovarian cancer, although cervical cancer risk seems unaffected^{68,69}.

Postmenopause, long-term morbidity and mortality. PCOS emerges as a risk factor for multimorbidity and chronic medication use, even at postmenopause^{34,63}. Although gynaecological manifestations of PCOS decline with age, metabolic issues and psychological distress, especially depressive symptoms, persist^{13,45,70}. Studies also reveal an increase in major adverse cardiovascular end points like myocardial infarction, stroke, heart failure and cardiovascular mortality beyond menopause^{12,35,49,71}, and hyperandrogenic status, at any point, seems to drive metabolic morbidity risks¹⁴.

Mortality data in PCOS is mixed^{72–74}, but a large register study reports increased overall mortality in PCOS⁵⁰, with contributions from cardiovascular outcomes and endometrial cancer.

Mechanisms/pathophysiology

Genetic factors

Family and twin studies provide robust evidence for the genetic underpinnings of PCOS⁷⁵. The correlation for PCOS between monozygotic twin sisters was twice as high as the correlation in dizygotic twins; heritability of PCOS was estimated at >70% in this twin study⁷⁶. Early candidate gene research provided limited replicable genetic associations⁷⁷. High-throughput genome-wide association studies (GWAS) have advanced our understanding, and initial GWAS in Han Chinese and European women identified 16 loci^{78–81} (Table 1).

Since then, an international consortium assembled data from >10,000 PCOS cases and 103,000 controls⁸². This GWAS meta-analysis identified 14 loci, of which three (*MAPRE1*, *ZBTB16* and *PLGRKT*) were novel. Regardless of diagnostic criteria (NIH, Rotterdam or self-reported (hyperandrogenism and irregular cycles)), 13 of the 14 loci were consistently associated with PCOS. Subsequent GWAS in electronic health record-linked biobanks identified one locus (*SOD2*)⁸³. In Northern Europeans, two independent rare variants in *CHEK2* and a common variant in *MYO10* were identified⁸⁴. Overall, several genes near loci discovered by GWAS implicate neuroendocrine (*FSHR*, *LHCGR* and *FSHB*) and metabolic dysfunction (*INSR*, *THADA* and *HMG2*) in the pathogenesis of PCOS, whereas the role of other genes remains to be identified. Of note, although loci are named based on genes in the vicinity, functional studies are needed to establish the effector gene at each locus. Studies in ovarian theca cells suggest that *PA2G4* and *ZNF217* may be the effector genes at the 12q13.2 and 20q13.2 loci, respectively, and could be a potential target for the treatment of ovarian dysfunction^{85,86}.

Large-scale GWAS results have enabled Mendelian randomization analyses to examine causality between PCOS and other phenotypes⁸⁷. Mendelian randomization utilizes genetic variants associated with an exposure to assess the causality of an exposure with an outcome without being affected by confounders that influence epidemiological studies. Mendelian randomization has suggested that high BMI^{79,82,88–90}, insulin resistance⁷⁹, increased fasting insulin⁵ and reduced levels of sex hormone-binding globulin (SHBG)^{6,79} may cause PCOS. However, another Mendelian randomization analysis did not find a causal role for PCOS in T2DM, coronary heart disease or stroke, suggesting that PCOS and cardiometabolic disease share common underpinnings (for example, obesity) rather than a causal relationship⁹¹. Moreover, PCOS seems to be associated with genetic variants that postpone menopause in women^{79,82}. Conversely, PCOS may increase the incidence of oestrogen receptor-positive breast cancer^{66,67} and decrease endometrioid ovarian cancer⁹².

The existence of PCOS subtypes has been proposed, including individuals with normal weight but an increased ratio of luteinizing hormone to follicle-stimulating hormone (FSH) and individuals with overweight or obesity with insulin resistance⁹³. A study performed unsupervised hierarchical cluster analysis of anthropometric, reproductive and metabolic traits in 893 individuals with PCOS⁹⁴ and identified a distinct ‘reproductive’ cluster with high luteinizing hormone and SHBG levels and a ‘metabolic’ cluster with high BMI, glucose and insulin levels. A GWAS stratified by BMI found shared as well as distinct genetic architecture in a lean PCOS subtype versus an obesity and overweight PCOS subtype⁹⁵. Delineating specific subtypes of PCOS in large prospective studies, with validation in independent cohorts, offers the potential for precision medicine in PCOS. Notably, one study indicated that men with high polygenic risk scores for PCOS had increased risk for developing obesity, T2DM, cardiovascular disease and male-pattern baldness, highlighting the impact of PCOS on both sexes⁹⁶ (Box 1).

Despite progress, the mechanistic details of how genetic variants contribute to PCOS, including the specific genes involved and their altered functions, remain unclear. A deeper understanding of these mechanisms is crucial for translating genetic discoveries into effective PCOS therapies.

Environmental and epigenetic factors

Non-genetic factors may also be involved in the pathophysiology of PCOS⁹⁷ and its inheritance could be attributed to genetic–epigenetic interactions⁹⁸ (Fig. 2). An aberrant maternal–fetal environment can contribute to epigenetic changes, potentially affecting fetal development. Such maternal factors could be weight gain during early pregnancy in women with PCOS⁹⁹ and sustained high levels of androgens and AMH throughout gestation¹⁰⁰, negatively impacting placental aromatase activity^{101,102}. Although exposure to environmental pollutants, including pharmaceuticals and personal care products, microplastics, endocrine disruptors, and nanoparticles, could potentially increase the risk of PCOS in the offspring, the evidence is still scarce, warranting further research¹⁰³.

Consistent with the Developmental Origin of Health and Disease hypothesis¹⁰⁴, offspring of mothers with PCOS exhibit elongated anogenital distance, an indicator of in utero androgen exposure¹⁰⁵. Daughters have a major risk of developing PCOS themselves, including reproductive and metabolic dysfunction^{106–108}, and sons have an increased likelihood of developing obesity and dyslipidaemia, suggesting in utero programming^{108,109} (Box 1). Global DNA methylation patterns in umbilical cord blood from neonates born to mothers with

Table 1 | Findings from GWAS in polycystic ovary syndrome

Locus	Index SNP	Nearest gene	First GWAS report	GWAS replication	Refs.
2p16.3 (A)	rs13405728	<i>LHCGR</i>	Chen, 2011	Shi, 2012	78,81
2p16.3 (B)	rs2268361	<i>FSHR</i>	Shi, 2012	Not applicable	81
2p21	rs13429458	<i>THADA</i>	Chen, 2011	Shi, 2012; Day, 2015; Day, 2018	78,79,81,82
2q34	rs1351592	<i>ERBB4</i>	Day, 2015	Day, 2018; Zhang, 2020; Tyrmi, 2022	79,82–84
5p15	rs9312937	<i>MYO10</i>	Tyrmi, 2022	Not applicable	84
5q31.1	rs13164856	<i>IRF1, RAD50</i>	Day, 2015	Day, 2018	79,82
6q25.3	rs17186366	<i>FNDC1, SOD2</i>	Zhang, 2020	Not applicable	83
8p23.1	rs804279	<i>GATA4, NEIL2</i>	Hayes, 2015	Day, 2018	80,82
9p24.1	rs10739076	<i>PLGRKT</i>	Day, 2018	Not applicable	82
9q33.3	rs2479106	<i>DENND1A</i>	Chen, 2011	Shi, 2012; Day, 2018; Tyrmi, 2022	78,81,82,84
9p24.1	rs4385527	<i>AOPEP, FANCC</i>	Shi, 2012	Hayes, 2015; Day, 2018	80–82
9p24.1	rs3802457	<i>AOPEP</i>	Shi, 2012	Not applicable	81
11p14.1	rs11031006	<i>FSHB</i>	Hayes, 2015	Day, 2015; Day, 2018; Tyrmi, 2022	79,80,82,84
11q22.1	rs1894116	<i>YAP1</i>	Shi, 2012	Day, 2015; Day, 2018	79,81,82
11q23.2	rs1784692	<i>ZBTB16</i>	Day, 2018	Tyrmi, 2022	82,84
12q13.2	rs705702	<i>RAB5B, SUOX</i>	Shi, 2012	Day, 2018	81,82
12q14.3	rs2272046	<i>HMGA2</i>	Shi, 2012	Not applicable	81
12q21.2	rs1275468	<i>KRR1</i>	Day, 2015	Day, 2018	79,82
16q21.1	rs4784165	<i>TOX3</i>	Shi, 2012	Day, 2018	81,82
19p13.3	rs2059807	<i>INSR</i>	Shi, 2012	Not applicable	81
20q13.2	rs6022786	<i>SUMO1P1</i>	Shi, 2012	Not applicable	81
20q11.21	rs853854	<i>MAPRE1</i>	Day, 2018	Not applicable	82
20q13.2	rs6022786	<i>SUMO1P1</i>	Shi, 2012	Not applicable	81
22q12	rs145598156	<i>CHEK2</i>	Tyrmi, 2022	Not applicable	84
22q12	rs182075939	<i>CHEK2</i>	Tyrmi, 2022	Not applicable	84

The index single-nucleotide polymorphisms (SNPs) listed are those from the first report of each locus. The SNPs in the above table are independent signals. Genome-wide association study (GWAS) replication indicates reports in which the same variant or a linked variant reached genome-wide significance.

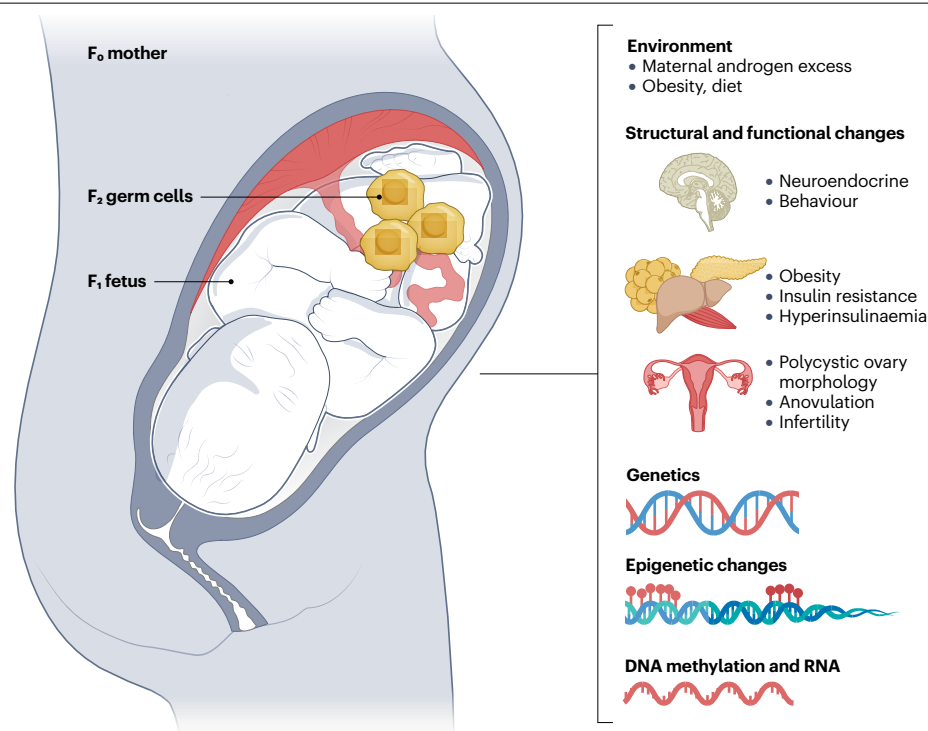


Fig. 2 | Genetic and epigenetic inheritance of PCOS. The inheritance of polycystic ovary syndrome (PCOS) likely involves genetic, environmental and epigenetic factors. F₀, the mother; F₁, first generation (somatic cells and germ cells); F₂, second generation (germ cells).

PCOS suggests a unique PCOS-associated epigenetic signature¹¹⁰, predisposing them to reproductive, metabolic and neuropsychiatric disorders^{97,106,109,111,112}.

Furthermore, studies have demonstrated the persistence of PCOS-like traits across generations in mice offspring not directly exposed (that is, third generation) to in utero dihydrotestosterone or AMH, suggesting epigenetic inheritance^{106,107,109}. Transcriptional and mitochondrial changes in oocytes¹⁰⁶, DNA methylation in ovaries¹⁰⁷ and small non-coding RNA alterations in sperm¹⁰⁹ accompany transgenerational transmission of reproductive and metabolic phenotypic traits in mice, suggesting that epigenetic modifications in germ cells may transmit PCOS across multiple generations. Importantly, some of the epigenetic and transcriptomic signatures are detectable in the serum of daughters and sons of women with PCOS, supporting the role of epigenetic inheritance in addition to genetic factors in the syndrome^{106,107,109}.

Hyperandrogenism

Elevated adrenal production of dehydroepiandrosterone and dehydroepiandrosterone sulfate, which through circulation reach peripheral target tissues and are then converted to oestrogens or androgens, can contribute to hyperandrogenism in women with PCOS^{113,114}. However, the characteristic hyperandrogenism – the primary reproductive phenotype of PCOS – results from excess ovarian androgen production (Fig. 3). The theca cells of the ovary are the primary site of production of androstenedione and testosterone and cells isolated from patients with PCOS show an intrinsic increase in androgen production in culture¹¹⁵. In vivo, luteinizing hormone and insulin are the primary stimulators of androgen production in combination with an absolute increase in the number of theca cells, which can be a consequence of increased follicle numbers and increased theca cell thickness¹¹⁶. A previously unrecognized gene

variant identified by GWAS, *DENND1A*⁷⁸, has been recognized in PCOS as a significant stimulator of the androgen pathway with several microRNA variants also implicated in the pathophysiology¹¹⁷. The ovarian granulosa cell layer is responsible for the aromatization of androgens to oestrogens under the influence of FSH, and most patients with PCOS exhibit high circulating levels of androgens, with consequences on multiple organ systems, including the skin, brain, adipose tissue and liver. Testosterone and androstenedione are the commonly elevated androgens and are amplified by the resultant low levels of SHBG, leading to elevated free androgens accessible to tissue androgen receptors¹¹⁸. Moreover, women with PCOS have insulin resistance with hyperinsulinaemia, which further decreases SHBG production in the liver and synergizes the effect of luteinizing hormone on theca cells, thereby contributing to elevated circulating levels of androgens¹¹⁹.

Ovarian function

The ovary contains many small follicles with oocytes that initially develop under the control of intra-ovarian growth factors and later under the influence of FSH following the development of appropriate receptors. Small (<9 mm) antral follicles produce large quantities of AMH (whereas large ones do not), which regulates the passage of follicular growth to preantral, larger follicles in association with androgen receptors preceding the expression of FSH receptors¹²⁰ (Fig. 3). High levels of androgens, AMH and paracrine regulators tend to prevent the spontaneous growth of antral follicles that can ovulate; however, this blockage can be overcome by exogenous FSH administration during fertility treatment. Elevated levels of follistatin and diminished concentrations of activin A have been proposed to potentially contribute to the arrest of follicular development (that is, not growing to >8–10 mm). This interplay may be partly responsible for the lack

of pre-ovular follicle development in PCOS¹²¹. Early luteinization can also occur under the influence of oestrogens and androgens and early development of the progesterone receptor prevents normal maturation of the preantral follicle¹²². The surrounding ovarian stroma is increased in PCOS and is highly vascularized, partially under the influence of the VEGF family, which in turn is upregulated by androgens¹²³. This phenomenon likely explains why ovarian hyperstimulation syndrome is more common in fertility treatment incorporating the use of exogenous FSH than in treatments that do not use FSH. Excess small antral follicles and high AMH can be normal physiological findings in normo-androgenic women, especially during adolescence¹²⁴, suggesting that additional stimuli to the ovary are required to fully develop PCOS.

Gonadotropic derangement

Gonadotrophin-releasing hormone (GnRH) is secreted in a pulsatile manner from the hypothalamus to the pituitary gland and concentrations of

luteinizing hormone and FSH are influenced by circulating oestradiol, progesterone and leptin¹²⁵. Kisspeptin neurons regulate GnRH neurons and play a crucial part in regulating GnRH pulse frequency and amplitude¹²⁶ (Fig. 3). Fast pulse frequencies favour luteinizing hormone secretion whereas slow frequencies favour FSH secretion¹²⁷. In PCOS, an increased pulse frequency of GnRH leads to high levels of luteinizing hormone in relation to FSH, thereby promoting the concept of a higher ratio of luteinizing hormone to FSH in women with PCOS of normal weight than in women without PCOS¹²⁸. By contrast, women with PCOS and obesity seem to have a luteinizing hormone-to-FSH ratio that is less affected. Given the absence of androgen receptors in GnRH neurons, it is plausible that elevated androgens disrupt the progesterone-induced negative feedback mechanism via a direct influence on kisspeptin neurons¹²⁶. Moreover, AMH receptors are also expressed in the hypothalamus, raising the possibility that circulating levels of this hormone may influence GnRH pulsatility¹²⁹. Another feature of PCOS is

Hypothalamus–pituitary–ovarian axis

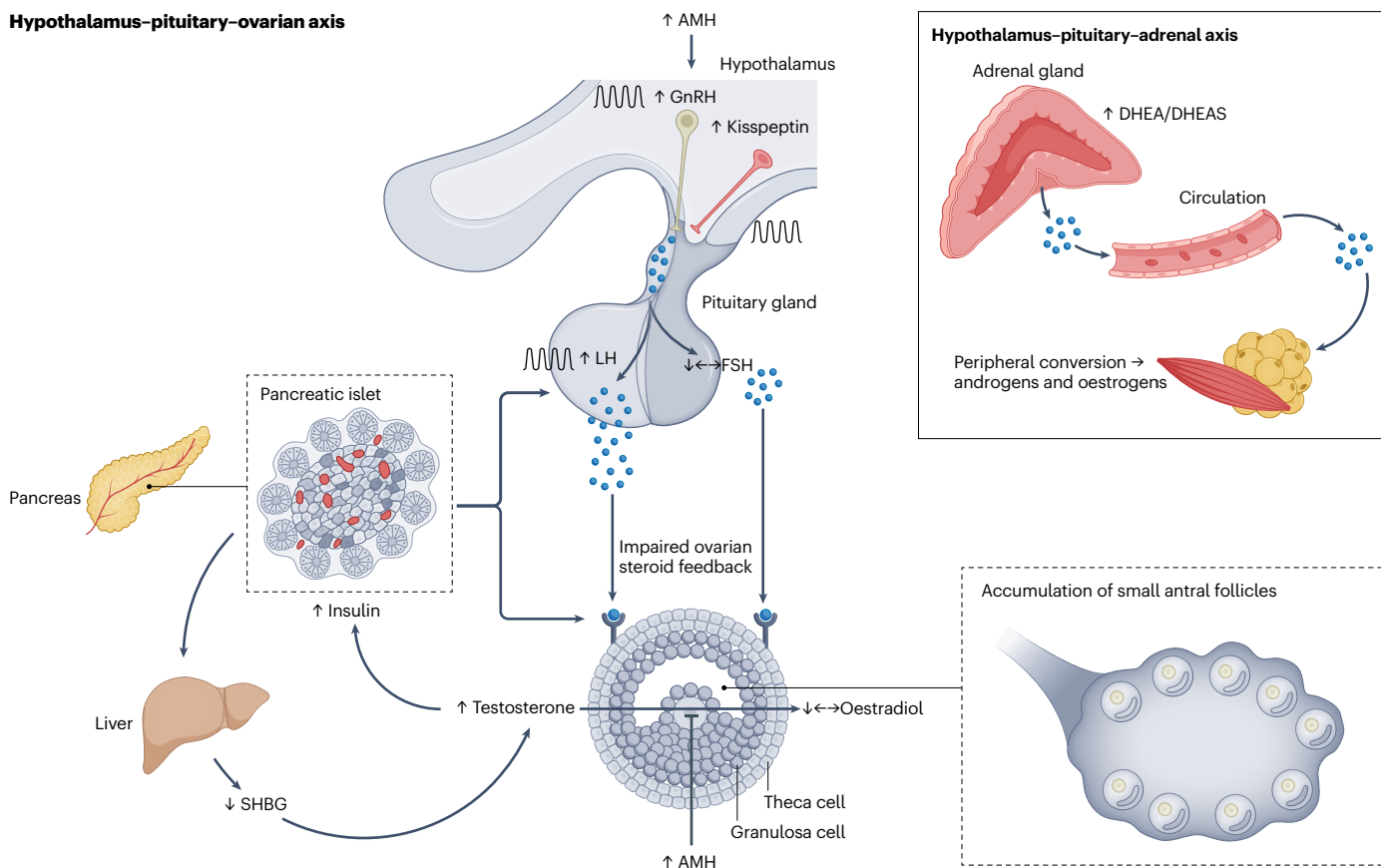


Fig. 3 | Alterations in the hypothalamic–pituitary–gonadal axis in women with PCOS. In women with polycystic ovary syndrome (PCOS), the gonadotropin-releasing hormone (GnRH) pulse frequency is elevated, increasing the luteinizing hormone (LH) frequency, whereas follicle-stimulating hormone (FSH) is unchanged or slightly reduced. GnRH secretion requires the stimulatory action of kisspeptin, produced by kisspeptin neurons located primarily in the arcuate nucleus of the hypothalamus. In PCOS, increases in kisspeptin likely trigger the increased GnRH pulse frequency, leading to increased LH pulse frequency. LH in turn stimulates theca cells to increase androgen production. Increased testosterone inhibits follicular growth, leading to the accumulation of many small antral follicles. Granulosa cells in the small antral follicles produce excessive

amounts of anti-Müllerian hormone (AMH). The function of AMH is to exert an inhibitory influence on the initial recruitment of follicles and the FSH-dependent progression and selection of preantral and small antral follicles <9 mm. The elevated AMH levels in women with PCOS inhibit the conversion of testosterone to oestradiol. Moreover, hyperinsulinaemia stimulates theca cells and suppresses sex hormone-binding globulin (SHBG) production, further contributing to hyperandrogenaemia. In addition, the adrenal gland produces excessive amounts of dehydroepiandrosterone (DHEA) and dehydroepiandrosterone sulfate (DHEAS), which via circulation reach peripheral organs (for example, adipose tissue, skeletal muscle and liver), where they are converted to oestrogens and androgens. Altogether, steroid feedback is impaired.

a degree of progesterone resistance to GnRH pulsatility, especially during puberty, which predisposes to ovarian androgen overproduction¹³⁰. These neuroendocrine changes in physiology offer opportunities for lifestyle changes and pharmaceutical interventions to restore normal pituitary function, thereby promoting ovarian follicular progression to ovulation.

Endometrial dysfunction

Endometrial dysfunction occurs in women with PCOS and anovulation owing to the absence of the corpus luteum, leading to low progesterone levels that compromise endometrial health over time¹³¹ (Fig. 4). Reduced progesterone allows unhampered oestrogen activity, increasing the expression of oestrogen receptor- α (ER α) and ER β , which promote endometrial cell proliferation¹³² and hyperandrogenism via increased endometrial androgen receptor expression¹³³. Hyperinsulinaemia and insulin resistance decrease the expression of insulin receptor and glucose transporter 4 (GLUT4)¹³⁴. Importantly, studies have shown compromised endometrial function during the secretory phase with altered steroid receptor expression and glucose transportation – factors that are known to be involved in delayed decidualization, impaired placentation, endometrial hyperplasia and potential carcinogenic processes^{131,135}. Moreover, reduced expression of uterine receptivity markers, for example, integrin, HOXA10 and IGFB1, and increased expression of inflammatory factors, such as TNF and NF- κ B, contribute to increased cell proliferation in PCOS^{134,136,137}. However, in analyses of receptivity markers in samples collected in the secretory phase, no discernible differences were observed between the endometrium of those with PCOS and that of those without PCOS with similar BMI¹³⁸.

Despite this finding, global gene expression profiling reveals distinct patterns in endometrial stromal cells from women with PCOS with increased expression of *SNCA*, *CDKN3*, and *CD58* and decreased expression of *PIBF1* and *MICB*¹³⁹, which are linked to endometrial cancer, endometriosis, delayed decidualization and increased inflammation¹⁴⁰. Moreover, exposing endometrial stromal cells to dihydrotestosterone disrupts gene expression associated with cell energy metabolism, proliferation, decidualization and inflammation¹³⁹. Only few studies exist on immune cell migration in the PCOS endometrium; however, the current data suggest some changes even in well-matched samples, thus warranting future studies^{141,142}.

Altogether, these findings underscore the multifaceted effect of hormonal and molecular changes in the endometrium of women with PCOS, potentially predisposing them to adverse reproductive outcomes such as early spontaneous miscarriage, pre-eclampsia and endometrial cancer^{143–146}. How specific cell types contribute to PCOS-specific endometrial dysfunction remains to be delineated, and these insights will help identify therapeutic targets (Fig. 4).

Insulin resistance

Women with PCOS, regardless of age, body composition and BMI, exhibit insulin resistance, characterized by a substantial reduction (35–40%) in insulin-stimulated glucose uptake, similar to that observed in individuals with T2DM^{147–149}. This insulin resistance is accompanied by compensatory hyperinsulinaemia, which strains pancreatic β -cells and reduces hepatic insulin clearance, in turn exacerbating hyperinsulinaemia¹⁵⁰.

Skeletal muscle dysfunction. The molecular basis of insulin resistance in PCOS is unique and tissue specific and involves increased

insulin-dependent serine phosphorylation of insulin receptors, impairing insulin signalling in skeletal muscle¹⁵¹, a distinguishing feature not present in other insulin-resistant conditions like obesity and T2DM¹⁵². Skeletal muscle insulin receptor substrate 1 (IRS1) is also increased¹⁵¹, leading to dysfunctional insulin signalling through the activation of MAPK pathways¹⁵³.

Furthermore, patients with PCOS exhibit abnormal DNA methylation and gene and protein expression profiles in skeletal muscle^{154–157}. These changes activate inflammatory response, lipid, glucose and fibrotic pathways, contributing to insulin resistance^{154–157}. Additionally, PCOS-related muscle dysfunction includes a reduction in insulin-sensitive type I muscle fibres and ectopic lipid accumulation¹⁵⁸ as well as impaired mitochondrial function¹⁵⁴, further aggravating insulin resistance (Fig. 5).

Despite these insights, the precise mechanisms and implications of PCOS on skeletal muscle dysfunction require further investigation, including understanding the roles of specific cell types in driving insulin resistance.

Adipose tissue dysfunction

Abdominal adipose tissue volume (including subcutaneous fat and visceral fat), measured with MRI is comparable between women with PCOS and those without PCOS, matched for BMI^{149,159}, and androgens exert no discernible impact on fat distribution¹⁶⁰. Nevertheless, distinctive morphological and functional abnormalities are evident within the adipose tissue of women with PCOS, with dysfunctional adipokine secretion resulting in increased circulating leptin and follistatin, and decreased adiponectin levels^{161,162}.

Enlarged adipocytes and decreased adiponectin levels, which are potent drivers of insulin resistance, are distinct features of PCOS¹⁴⁹. Moreover, hyperandrogenism correlates with adipocyte hypertrophy¹⁶³, inhibits adipose stem cell commitment to preadipocytes¹⁶⁴ and hinders their differentiation into mature adipocytes¹⁶⁵. In addition, differentiated adipocytes from women with PCOS have increased lipid content than those from women without PCOS, correlating with androgen levels^{166,167}.

PCOS-specific adipose tissue dysfunction extends to impaired insulin-stimulated glucose uptake, which is attributed to decreased GLUT4 expression^{168,169}. Moreover, subcutaneous adipocytes of women with PCOS with normal weight are resistant to catecholamine-induced lipolysis due to a decreased number of β 2-adrenergic receptors^{170,171}. Together with diminished hormone-sensitive lipase and protein kinase A (PKA) regulatory components, altered lipolysis is a possible link to insulin resistance^{170,172–174} (Fig. 6). Additionally, adipocytes in PCOS exhibit dysfunctional lipid storage, release and energy regulation^{175,176}.

Increased intra-adipose production of testosterone and dihydrotestosterone in PCOS via increased expression of androgen-activating enzyme aldo-ketoreductase 1 C3 (AKR1C3) drives lipid accumulation, resulting in systemic lipotoxicity¹⁷⁷ with ectopic lipid deposition in tissues such as skeletal muscle and liver¹⁵⁸, leading to insulin resistance. Inflammation, macrophage accumulation, fibrosis and adipose tissue expansion further exacerbate insulin resistance in PCOS^{178,179} (Fig. 5).

These findings underscore the specificity of adipose tissue dysfunction in women with PCOS. Although various global protein, transcriptional and epigenetic changes may contribute to PCOS-specific dysfunctions^{158,180,181}, the precise roles of specific cell types within adipose tissue, their molecular functions and inter-tissue communication are yet to be defined.

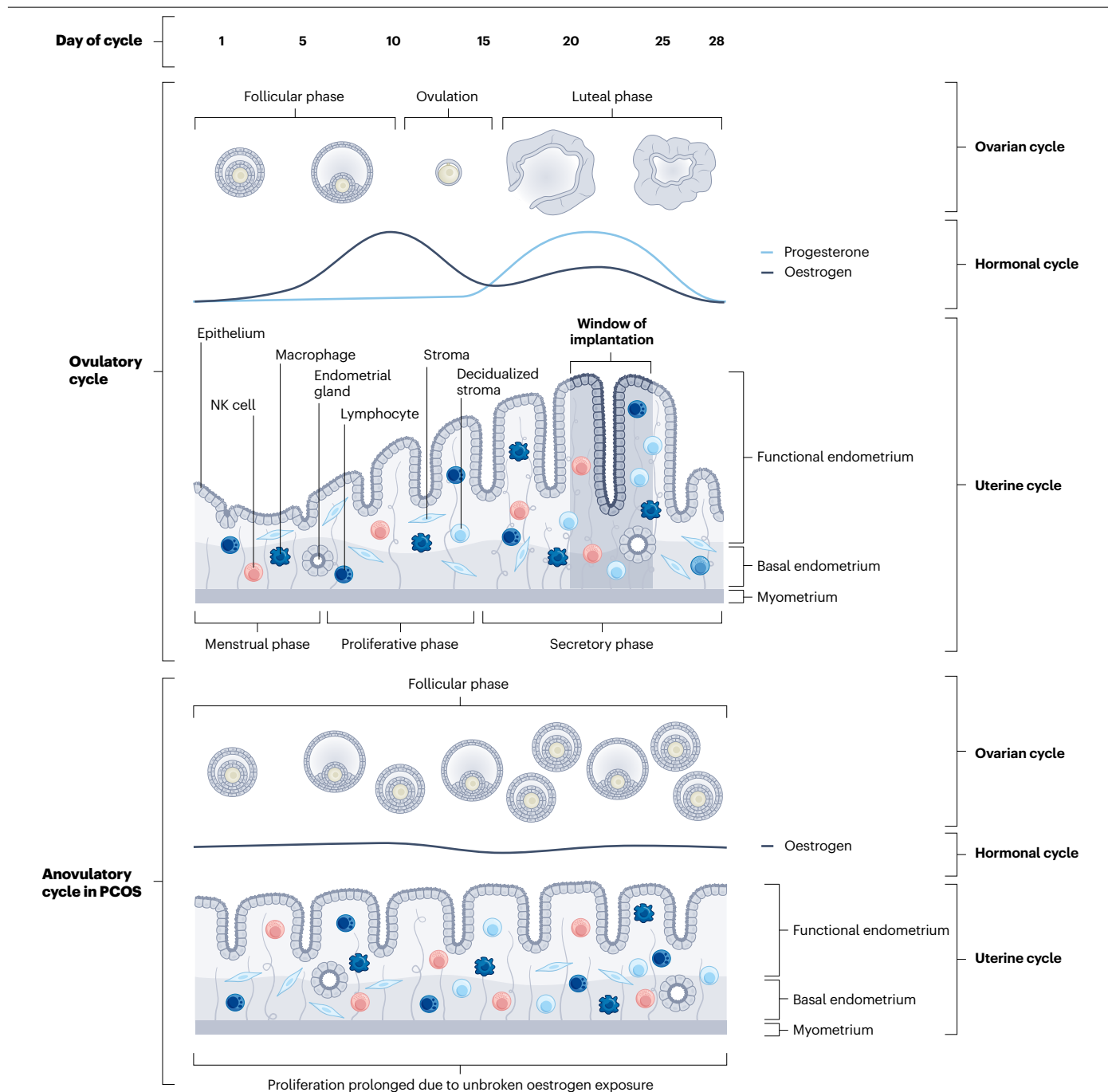


Fig. 4 | The normal endometrium and the anovulatory endometrium. Endometrial development requires follicular maturation in response to ovarian steroid hormone secretion (oestradiol and progesterone). In women with polycystic ovary syndrome (PCOS) and irregular cycles, the endometrium is exposed to oestrogen without counteracting progesterone action. The endometrium of women with PCOS is characterized by more epithelial cells and

less stromal cells compared with the normal cycling endometrium. Endometrial proliferation is linked to increased expression of oestrogen receptor- α (ER α) and ER β as well as androgen receptor. Insulin resistance negatively affects the expression of insulin receptor and glucose transporter 4 (GLUT4). NK, natural killer.

Altered immune function and inflammation

Comorbidities in PCOS are linked to altered immune function, implying low-grade inflammation¹⁸². Numerous studies have reported elevated

circulating cytokines, such as TNF, IL-6, IL-18, MCP1 and high-sensitivity CRP, in women with PCOS^{183–185}, indicating a systemic pro-inflammatory status. These cytokines may originate from peripheral blood

leukocytes, with increased levels of lymphocytes, monocytes, natural killer cells and neutrophils in PCOS¹⁸⁶.

Cytokines are also produced in specific cells within tissues such as adipose tissue, endometrium and ovaries. Subcutaneous adipose tissue in women with PCOS shows elevated expression of CD11c and TNF, along with an increased density of crown-like structures, suggesting altered macrophage activity¹⁷⁸ and elevated CCR5 in omental fat¹⁸⁷; however, another study found no differences in CD68⁺ macrophage density¹⁴⁹. The PCOS endometrium exhibits a dysfunctional immune profile, with an increased proportion of specific immune cell populations, including macrophages, and increased expression of markers such as TNF, IL-6 and NF- κ B^{136,142}, whereas studies on ovarian immune profile remain inconclusive¹⁸².

Autoimmune components are implicated in PCOS, with reported autoantibodies targeting the thyroid and GnRH receptor¹⁸⁸. Moreover, B cell numbers are increased in PCOS, potentially contributing to increased inflammation and autoantibody production¹⁸⁹, although research suggests that B cells may not be central mediators of PCOS pathology¹⁹⁰.

In summary, PCOS is associated with low-grade inflammation (Fig. 5), but its impact on disease progression and comorbidities remains unclear and warrants further investigation.

Cardiovascular dysfunction

Studies have reported structural alterations, such as increased left ventricular mass and increased left atrial diameter¹⁹¹, endothelial dysfunction¹⁹², coronary and aortic calcification¹⁹³, and increased intima-media thickness of common carotid and femoral arteries¹⁹⁴, in women with PCOS, especially in those with insulin resistance and

hyperandrogenaemia. Moreover, increased levels of fibrinogen and plasminogen activator inhibitor 1 (PAI1) suggest a prothrombotic state in women with the syndrome¹⁹⁵. Together, these alterations may contribute to their increased risk of cardiovascular disease and mortality. In addition, cardiovascular dysfunction is associated with elevated muscle sympathetic nerve activity¹⁹⁶. Of note, women with PCOS have increased muscle sympathetic nerve activity independent of BMI^{197–199} and exercise. Electrical stimulation and renal denervation not only decrease muscle sympathetic nerve activity but also improve hyperandrogenism, insulin resistance and hypertension^{200–204}. Yet, whether the mechanisms leading to impaired cardiac autonomic function in women with PCOS rely on direct or indirect effects of hyperandrogenism, hyperinsulinaemia and insulin resistance²⁰⁵, arterial stiffness²⁰⁶, and/or lower baroreflex sensitivity with lower heart rate variability remains to be determined^{207,208}. Understanding the underlying mechanisms driving increased sympathetic output is critical as these could be promising therapeutic targets in women with PCOS.

Liver dysfunction

Obesity together with impaired glucose metabolism, increased triglyceride and cholesterol levels, and hyperandrogenism are major risk factors for the development of MASLD, a condition where hepatocytes accumulate triglycerides owing to macrovesicular steatosis⁵¹. In women, high circulating testosterone levels increase lipid storage in the liver and contribute to impaired glucose metabolism and development of T2DM⁵¹. Moreover, hyperandrogenism in women with PCOS is linked to reduced hepatic SHBG production, which is suggested to be a biomarker of MASLD in PCOS²⁰⁹. Liver dysfunction plays a central part in

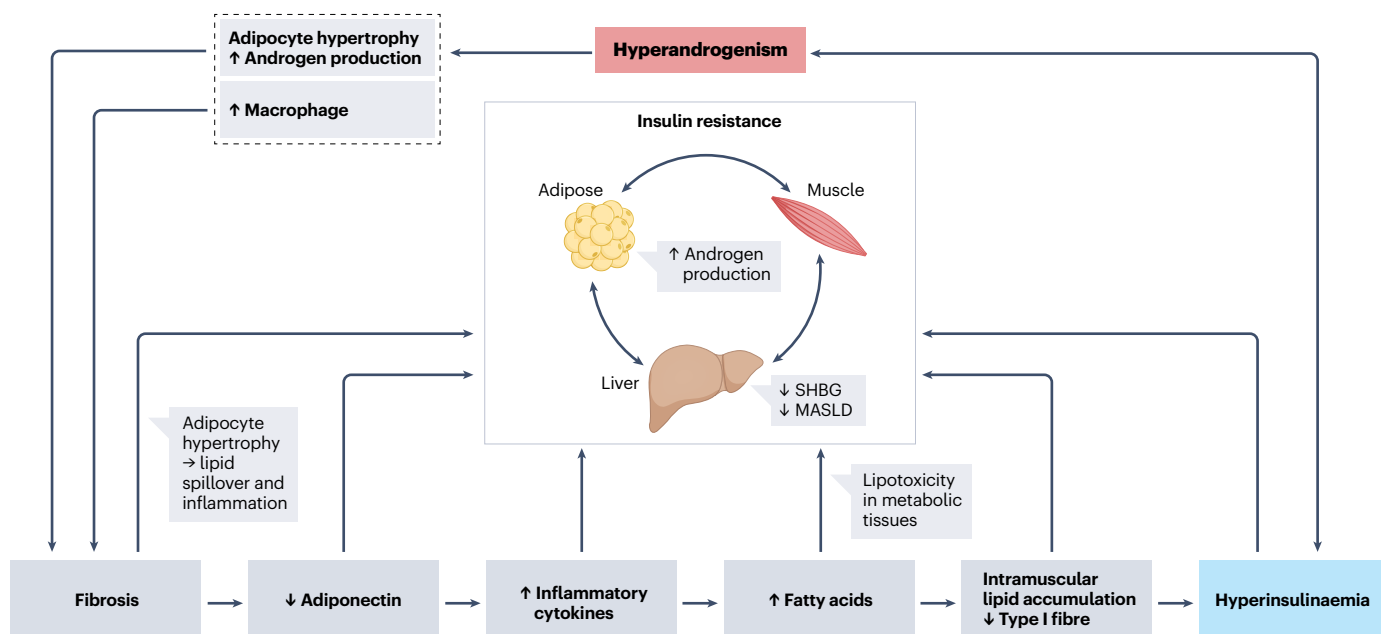


Fig. 5 | Proposed vicious circuit triggered by hyperandrogenism contributing to insulin resistance in PCOS. Hyperandrogenism and hyperinsulinaemia are two key features of polycystic ovary syndrome (PCOS), driving insulin resistance in the adipose tissue, skeletal muscle and liver. The ovaries not only produce excessive amounts of androgen but also the adipose tissue contributing to adipocyte hypertrophy and infiltration of macrophages leads to systemic inflammation with an increased release of cytokines and a decrease in

adipokines such as adiponectin. Moreover, adipocytes cannot store lipids and a lipid spillover occurs that contributes to lipid accumulation in other tissues, including liver and skeletal muscle, leading to lipotoxicity. This leads to tissue-specific insulin resistance and tissue crosstalk. MASLD, metabolic dysfunction-associated steatotic liver disease; SHBG, sex hormone-binding globulin. Adapted from ref. 290.

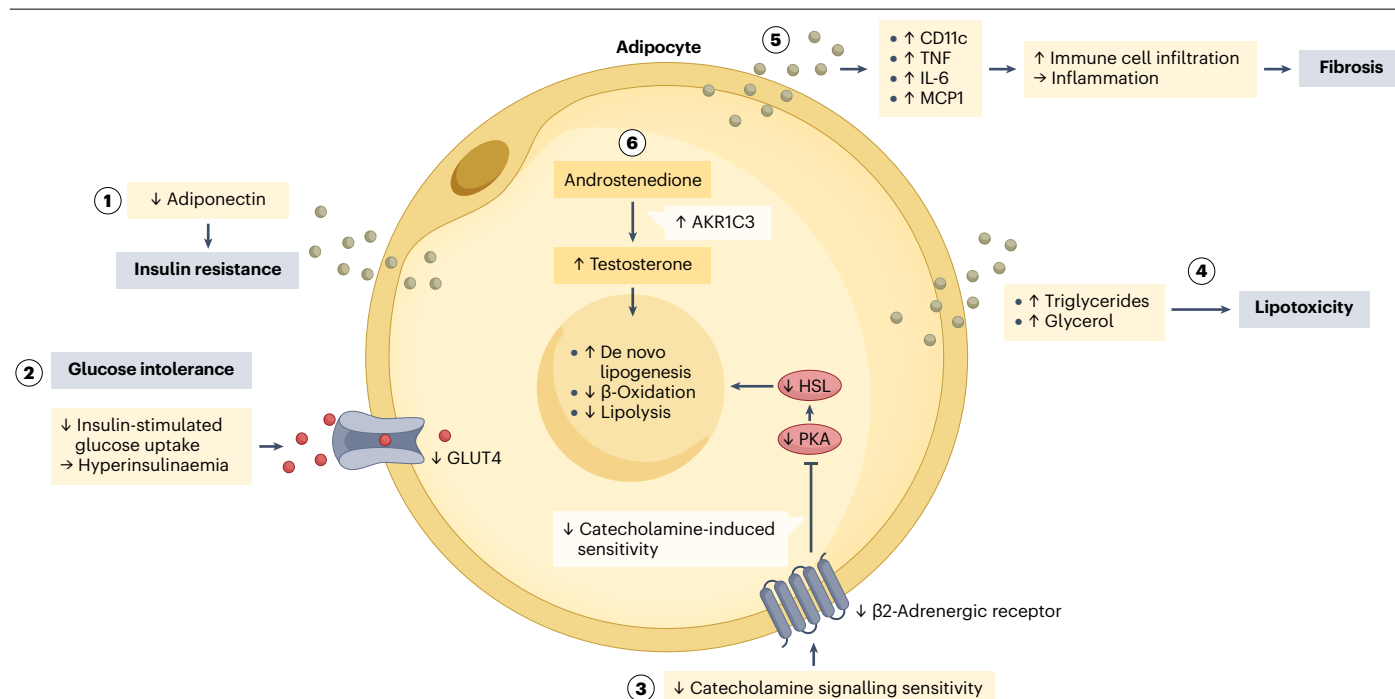


Fig. 6 | Proposed mechanisms of adipose tissue dysfunction in PCOS.

(1) Adipocyte hypertrophy and decreased production and secretion of adiponectin are strong drivers of insulin resistance in polycystic ovary syndrome (PCOS), independent of obesity. However, current knowledge on insulin signalling and lipolysis in subcutaneous adipose tissue in PCOS is limited. (2) Insulin resistance with reduced glucose transporter 4 (GLUT4) translocation and glucose uptake results in glucose intolerance. Moreover, the lipolysis in PCOS adipose tissue is altered in both basal and catecholamine-stimulated conditions. (3) Catecholamine signalling sensitivity is reduced and the overall lipolytic stimulation in subcutaneous adipocytes is blunted in PCOS. Although many steps in the stimulation pathway are not yet explored in PCOS, part of the blunted response can be attributed to decreased levels of β2-adrenergic

receptors as well as to protein kinase A (PKA) resulting in lower phosphorylation of hormone-sensitive lipase (HSL). (4) Altered lipolysis results in altered lipid storage and increases in circulating triglycerides and glycerol, contributing to lipid accumulation in other tissues and lipotoxicity. (5) Moreover, increased cytokine secretion results in infiltration of immune cells, contributes to low-grade inflammation and is linked to fibrosis. (6) Finally, specific for PCOS is the increased androgen production driven by increased expression of androgen-activating enzyme aldo-ketoreductase 1 C3 (AKR1C3), which converts androstenedione to testosterone (and dihydrotestosterone), increasing de novo lipogenesis and decreasing β-oxidation and lipolysis. This further drives ectopic lipid accumulation, lipotoxicity, fibrosis and insulin resistance. Adapted from ref. 37.

regulating the immune system, with accumulation of lipids in the liver leading to lipotoxicity, which in turn stresses hepatocytes to generate inflammatory factors triggering the immune system²¹⁰. Although not yet studied, lipotoxicity and MASLD likely trigger the immune system and contribute to the observed low-grade inflammation in women with PCOS but need further investigation (Fig. 5).

Psychological disorders

The aetiology and pathophysiology of psychiatric disorders in women with PCOS likely involve a complex interplay of genetic and environmental factors⁹⁸. Twin and familial studies suggest substantial heritability of psychiatric disorders²¹¹, and Mendelian randomization analyses indicate a causal link between PCOS and depression⁸². However, current research has found no genetic correlation between PCOS and psychiatric disorders²¹², and the association between depression and PCOS weakens when BMI is taken into account, emphasizing the role of obesity.

Non-genetic factors, such as an adverse maternal–fetal environment owing to elevated androgens, can adversely impact fetal brain development, leading to anxiety-like behaviour in the offspring^{111,213,214}. Experimental rodent data indicate that androgen exposure during

critical brain development stages²¹⁵ can mediate anxiety-like behaviour via various factors, for example, ERα, serotonergic signalling, GABAergic signalling and activation of adrenoceptor α1B, in the amygdala and hippocampus^{213,214}. In support of preclinical observations, maternal PCOS is suggested to be linked to increased autism spectrum disorder in offspring of either sex²¹⁶ and an increased risk of attention-deficit/hyperactivity disorder in boys only²¹⁷. Epidemiological studies also suggest associations between maternal PCOS and various psychiatric disorders, such as attention-deficit/hyperactivity disorder, and autism spectrum disorder in children^{97,218,219}. Moreover, anxiety disorders in mice can be transgenerationally transmitted in both sexes¹¹¹.

Psychosexual dysfunction. Additionally, women with PCOS often experience psychosexual dysfunction, which may be related to external factors such as body image concerns; however, the precise underlying mechanism remains unknown^{62,220,221}. Of note, adult women with PCOS and a mother with the diagnosis display impaired sexual function, indicating that maternal PCOS may trigger not only psychiatric disorders but also sexual dysfunction²²². Moreover, a preclinical mouse model of PCOS suggests a potential link between sexual behaviour impairment

in female offspring and decreased hypothalamic expression of progesterone-sensitive neuronal nitric oxide synthase²²³.

Diagnosis, screening and prevention

Diagnosis

Since the 2000s, PCOS diagnosis has changed as the understanding of the condition has evolved from a crude description of macroscopic ovarian changes to a complex endocrine condition caused by polygenic, epigenetic and environmental factors²²⁴. Ultimately, in 2018, building on the American Society for Reproductive Medicine and the European Society of Human Reproduction and Embryology Rotterdam consensus criteria²⁵, an international PCOS network with 40 organizations reaching across six continents upgraded these criteria from consensus to the International Evidence-based Guideline Diagnostic Criteria²²⁵ (Fig. 7).

Changes to the International Guideline included a robust development process with patient and health professional groups; comprehensive evidence synthesis and refined individual diagnostic components, including increasing the cut-off for arrested ovarian follicle count (<9 mm) from >12 to ≥20 per ovary and/or the cut-off for enlarged ovaries to >10 cm³ (ref. 225) (Fig. 7). Separate criteria for adolescents were developed requiring the presence of both oligoanovulation and hyperandrogenism, excluding ultrasound in the diagnostic criteria of adolescents owing to a lack of specificity for PCOS²²⁶. Adolescents 'at risk' were recognized when one of the two aforementioned features was present, with reassessment recommended once they reach adulthood. These altered criteria for adolescents were validated in a well-characterized longitudinal cohort and dramatically reduced overdiagnosis at this life stage²²⁷. Diagnosis in adults and adolescents requires biochemical exclusion of thyroid dysfunction, hyperprolactinaemia, congenital adrenal hyperplasia, primary and secondary ovarian failure, and clinical exclusion of hypogonadotropic hypogonadism, as well as of other less common causes such as adrenal carcinoma or Cushing disease.

The International Guideline is updated every 5 years and, in 2023, the evidence in PCOS was reviewed, synthesized, and appraised for quality and integrity and a 6,000-page technical report was presented with improved evidence certainty and strength of recommendations²⁴. Diagnostic criteria were expanded to include either ovarian ultrasound or AMH levels in adults only^{1,24} (Fig. 7). This update is now a living guideline, responsive to the rapidly evolving advances in research, called the 'International Guideline diagnostic criteria'.

Another key advance in knowledge around diagnosis relates to subgroups nested under the umbrella of PCOS. Traditionally defined as phenotypes A–D, based on Rotterdam diagnostic criteria^{25,224}, these original crude groups do not align to aetiology, genotype or pathophysiology. Exploration of GWAS combined with sophisticated cluster analyses in data sets has suggested grouping around core endocrine features across hyperandrogenaemia, hyperinsulinaemia, hypothalamic–pituitary abnormalities and BMI^{19,94,224,228}. Reproductive, metabolic and indeterminant subgroups have emerged and progress in understanding the underlying aetiology and pathophysiology is likely to nuance these distinctions further, similar to T2DM. T2DM has a similar complex polygenic aetiology to PCOS and extensive subgroups have been identified, including many subtypes of maturity-onset diabetes of the young²²⁹ and proposed subtypes of adult-onset diabetes.

Evolving diagnostic criteria and lack of clarity on the subtypes, alongside inadequate health professional education and patient information, contribute to delayed PCOS diagnosis and patient dissatisfaction¹⁸. Delayed diagnosis also limits screening, prevention and treatment of the diverse features of PCOS, with health implications.

Screening for PCOS and prevention opportunities can improve outcomes, including reproductive health (Box 2).

Management

Metabolic dysfunction

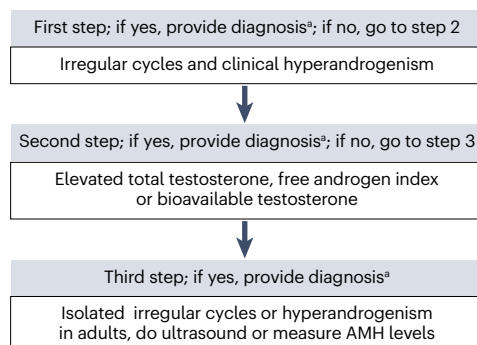
Lifestyle modification. Lifestyle interventions are a cornerstone in managing PCOS, particularly in individuals with excess weight. Modest reductions in total body weight, as little as 5%, offer metabolic and possibly reproductive benefits^{57,230–232}. A meta-analysis of 18 randomized controlled trials (RCTs) demonstrated that lifestyle interventions improve various parameters, including waist circumference, waist-to-hip ratio, modified Ferriman–Gallwey score (which evaluates hirsutism), fasting insulin, total cholesterol and LDL cholesterol compared with minimal intervention in PCOS²⁴. Although RCTs investigating the effect of lifestyle interventions in the general population suggest a prevention of T2DM in individuals at risk²³³, similar trials in PCOS are yet to be performed. Notably, studies involving lifestyle interventions resulting in weight loss demonstrated that women with PCOS achieve similar weight loss improvements as those without PCOS²⁴.

Notably, diet composition, including macronutrient components, does not significantly impact weight loss or metabolic outcomes in women with PCOS or the general population in RCTs^{234–236}. Currently, no optimal lifestyle intervention exists for PCOS, with general population guidelines applicable in this condition, tailored to personal preference²³⁷. Similarly, intensive exercise interventions alone do not yield superior weight loss or metabolic improvements in PCOS²⁴. Overall, evidence supporting any specific type or intensity of exercise for anthropometric, metabolic, hormonal, reproductive or psychological outcomes is lacking^{238,239}, although engaging in 150 min of moderate-to-vigorous exercise per week is recommended for overall cardiometabolic risk improvement.

Bariatric surgery. In individuals with BMI >35 kg/m², lifestyle interventions are not as effective over the long term to maintain improvement in weight and metabolic disease. In the general population, bariatric surgery or metabolic surgery has been demonstrated to provide substantial and sustainable weight loss with accompanying improvement in metabolic health^{240,241}. In a non-RCT comparing surgery, oral contraceptive and metformin management of PCOS, the greatest improvement in metabolic outcomes was observed with surgery, with regards to decreased weight, BMI, waist circumference, hip circumference, fasting glucose, fasting insulin, triglycerides, LDL, haemoglobin A1c (HbA1c), total testosterone and free androgen index²⁴². In studies comparing surgical outcomes between women with and without PCOS, similar improvements were observed in both groups concerning total weight loss, total testosterone, SHBG, AMH, total cholesterol, LDL, triglycerides or HDL^{242,243}, although improvement in fasting glucose levels and BMI post-surgery was greater in women with PCOS than in women without PCOS^{242,243}.

Medical treatment. Although no drugs are approved specifically for the medical management of metabolic dysfunction in PCOS, metformin, a biguanide used to treat T2DM, has been used extensively in the management of PCOS. In RCTs investigating metformin and placebo, metformin significantly improved HOMA-IR and fasting glucose and improved waist-to-hip ratio and lipid profiles²⁴⁴. In a meta-analysis of RCTs conducted in women with PCOS, use of glucagon-like peptide 1 (GLP1) receptor agonists, such as liraglutide, yielded modest metabolic improvements over a 12-week period compared with metformin alone.

a



b

Adolescents Fulfil two

In adolescents, hyperandrogenism and ovulatory dysfunction are required

Clinical or biochemical hyperandrogenism

- Hirsutism: Ferriman–Gallway score ≥ 4 –6 or
- Total or free testosterone measures by LC–MS or calculated free androgen index



Ovarian dysfunction

- Irregular menstrual cycles and/or anovulation: < 21 days or > 35 days with > 3 years after menarche



Polycystic ovarian morphology or AMH

- Antral follicles: ≥ 20 (2–9 mm) or
- Enlarged ovaries: > 10 cm³ or
- AMH



**Adult
Fulfil two out of three**

Fig. 7 | Diagnosis of PCOS. a, The first step includes presence of irregular cycles and clinical hyperandrogenism such as hirsutism and/or female pattern hair loss; if yes, polycystic ovary syndrome (PCOS) is diagnosed. The second step involves the performance of biochemical testing of circulating elevated total testosterone, calculation of free androgen index, or bioavailable testosterone; if levels are increased, PCOS is diagnosed. In the third step, if isolated irregular cycles or hyperandrogenism are reported, ultrasound is performed to assess antral follicle count (> 20 that are < 9 mm) or for enlarged ovary (> 10 cm³) or elevated circulating anti-Müllerian hormone (AMH) levels. Importantly, the third step is not performed in adolescents. **b**, Diagnostic criteria for PCOS in adolescents (left) and in adult women (right). LC–MS, liquid chromatography–mass spectrometry. ^aAfter exclusion of other causes.

However, combination therapy with metformin demonstrated more substantial improvement in weight reduction, fasting glucose and fasting insulin than either treatment alone²⁴⁵.

The use of medications targeting weight loss alone in PCOS remains uncertain although lifestyle interventions alone may prove insufficient. Metformin has shown modest weight loss benefits and improved body composition compared with oral contraceptives alone or in combination in women with PCOS²⁴⁶. Several anti-obesity drugs, particularly GLP1 receptor agonists, are approved for use in adults but face cost, access and availability challenges; anti-obesity drugs are not recommended when pregnancy is desired.

Studies have explored the use of GLP1 receptor agonists, such as liraglutide and semaglutide, for weight loss and metabolic improvements in PCOS. In RCTs, liraglutide led to weight and fat mass reduction²⁴⁷. However, liraglutide alone resulted in more lean body mass loss than placebo, which improved when combined with lifestyle interventions²⁴⁷. A non-randomized study with semaglutide also showed weight loss and some metabolic improvements in women who had previously undergone lifestyle modifications without success, although not across all parameters²⁴⁸. Although based on a limited number of small studies, GLP1 receptor agonists are shown to reduce free androgen index²⁴⁷ and to normalize menstrual cycles²⁴⁸; large trials are needed in the future. Of note, gastrointestinal side effects are common with GLP1 agonists as well as weight rebound after cessation.

Inositol seems to have limited value in treating metabolic outcomes in PCOS, with metformin being more effective. However, inositol could be used with informed consent and where access to metformin or more effective medications is limited²⁴.

Hyperandrogenism

Suppression of ovarian androgen secretion. In PCOS, common androgen-related features include hirsutism, acne and female pattern hair loss¹. Low-dose ethinyl oestradiol combined oral contraceptive pills (COCPs) are the preferred treatment for hirsutism in both adults and adolescents as they increase SHBG levels and decrease luteinizing hormone, serum testosterone and androstenedione levels²⁴⁹. Meta-analysis shows that cyproterone acetate and fourth-generation progestins result in greater reduction in serum androgen levels than earlier progestins²⁵⁰. However, fourth-generation progestins demonstrated no differences in their effect on hirsutism²⁵⁰. Given the increased risk for venous thrombosis associated with ethinyl oestradiol plus cyproterone acetate, these preparations are considered second-line therapy¹. Addressing the primary symptoms and considering cosmetic therapies are crucial. The choice of treatment should involve shared decision-making, addressing contraindications and side effects.

Combining COCPs with lifestyle management is more effective than COCP–metformin combinations¹. Although metformin and COCPs can improve hyperandrogenism, a meta-analysis showed that COCP treatment results in lower free androgen index, total and free testosterone, androstenedione, and dehydroepiandrosterone sulfate levels along with higher SHBG levels than metformin treatment; however, hirsutism did not differ between the treatments²⁵¹.

Anti-androgens. Anti-androgens could be considered to treat hirsutism in women with PCOS if suboptimal response is observed after a minimum of six months of COCP and/or cosmetic therapy²⁴.

If anti-androgens are used, the side effects should be weighed against the possible benefits considering the extent and the impact of hirsutism on the individual patient.

Bariatric surgery could also be considered to improve hirsutism. One large study reported a complete resolution of hirsutism after bariatric surgery in 75% of all patients at the end of 6 months of follow-up²⁵². Another 20% experienced a moderate resolution whereas the remaining 5% showed limited or no resolution of the extent of their hirsutism.

Cosmetic treatments. Mechanical laser and light therapies should be considered to reduce facial hirsutism and related depression, anxiety, and quality of life in women with PCOS. A greater number of laser treatment sessions may be required in women with PCOS than in women with idiopathic hirsutism to achieve hair reduction²⁴. Adverse effects seem limited in the hands of experienced and suitably qualified providers and women should be encouraged to seek hair reduction therapies from such providers. Wavelength and delivery of laser treatment varies by skin colour and hair colour. Moreover, laser is relatively ineffective in women with blond, grey or white hair. Finally, the addition of COCPs, with or without anti-androgens, to laser treatment may provide greater hair reduction and maintenance than laser alone¹.

Endometrial protection

Progestin-containing regimens offer endometrial protection. In PCOS, COCPs are commonly used as they are considered first-line therapy to manage irregular cycles¹. Most COCPs have continuous progestin dosing, effectively controlling endometrial proliferation and offering

cycle regulation. Progestin-only preparations can also be considered, although they lack the beneficial effect of COCPs to control hyperandrogenism. Levonorgestrel intrauterine devices are effective for endometrial protection, although some women may suffer from extrauterine levonorgestrel effects such as acne and hair loss²⁵³. Cyclic progestins, like medroxyprogesterone or norethisterone, are also used, usually for 10–14 days to induce menstruation over 8–6 cycles per year. Different weight loss strategies and metformin also support endometrial health and should be considered in PCOS^{131,254}. In cases where hyperplasia or cancer are suspected, endometrial biopsy is indicated.

Fertility

Medication. Fertility treatment in women with PCOS should start with preconception counselling and optimizing maternal health before pregnancy²⁵⁵. Preconception weight loss through lifestyle interventions or bariatric surgery may improve ovulation, enhance natural pregnancy rates^{242,256}, and may increase ovulation rates with first-line treatments²⁵⁷. However, pregnancy in PCOS carries risks, including miscarriage, gestational diabetes, pregnancy-induced hypertension, pre-eclampsia, preterm delivery and the need for a Caesarean section^{1,258}. The fetus is at risk for intrauterine growth restriction, with small-for-gestational-age babies and low birthweight at birth^{1,258,259}. Given the elevated risks, the primary goal of fertility therapy should be a singleton pregnancy to avoid complications associated with multiple pregnancies.

The first-line treatment for anovulation in PCOS involves ovulation induction with letrozole over several cycles, with dose escalation if anovulation persists^{1,260–262}. Initial concerns about possible

Box 2

Screening and prevention

Infertility, time to conception and inter-conception intervals, rate of fertility treatment and maternal age are all increased in women with polycystic ovary syndrome (PCOS), with lower fecundity rate and fewer children than in women without PCOS^{54,56}. A delayed diagnosis of PCOS adversely impacts fertility and family size and increases advanced maternal age in a condition where pregnancy complications are already increased. In this context, a reproductive life plan is recommended to focus on timely diagnosis, screening, prevention and family planning¹. On the other hand, women should be informed about the chance of spontaneous pregnancies and that they are likely to have at least one child; however, this may require assisted reproductive technologies^{55,56,303}.

Preconception offers additional screening opportunities. Nutritional status, prenatal vitamin use, smoking, alcohol, diet, exercise, sleep, and mental, emotional and sexual health as well as screening for comorbidities such as excess weight, type 2 diabetes mellitus and high blood pressure, are important to improve maternal–fetal health and overall health. Risks for gestational diabetes and hypertension during pregnancy are increased in women with PCOS⁵⁸. Screening, monitoring and prevention of complications in pregnancy are strongly recommended in the International Guideline, including identifying PCOS status, high BMI, excess gestational weight gain, hypertension and type 2 diabetes mellitus, with a focus on improving maternal–fetal outcomes^{1,58}.

For endometrial protection, monitoring cycle regularity for those not on hormonal contraception is important to limit hyperplasia and endometrial cancer, especially in those with a high BMI. However, the absolute risk for cancer remains low and routine screening is not recommended unless clinically indicated^{1,24}.

Given that anxiety and depressive symptoms are increased in 70–80% of those with PCOS, screening for depression and anxiety are routinely recommended, using regionally validated and assessable tools. For eating disorders, screening is not routinely recommended; however, where these disorders are suspected, screening is recommended^{1,24}. Other psychological features, including body image distress and self-esteem, should be considered for screening and treatment if clinically indicated. Individualized screening, identification and treatment of psychological features should be a priority to improve quality of life^{1,24}. The frequency of psychological screening should be based on clinical assessment and considered when other intersectional factors or life events occur such as infertility or pregnancy³⁰⁴.

Cardiometabolic screening is recommended routinely in PCOS from the time of diagnosis. Weight, glucose levels, blood pressure and lipid profiles should be routinely and regularly screened to prevent complications^{1,24}.

teratogenicity have been allayed by a systematic review noting similar rates of birth defects as other ovulation induction medications, although use of letrozole is still off-label²⁶³. Other oral medications, in order of reduced effectiveness, include the combination of metformin and clomiphene citrate, clomiphene citrate alone, and metformin alone²⁶¹. Pregnancy rates often require six or more cycles with most medications, and the risk of multiple pregnancies ranges from 3% to 7%²⁶⁴. Some guidelines suggest low-dose gonadotropin therapy as a second-line treatment but careful management is needed to prevent multiple follicle recruitment and ovarian hyperstimulation. Continued use of oral medications with clomiphene citrate may also be effective in the long run²⁶⁴.

IVF and elective single embryo transfer. In vitro fertilization (IVF) has gained popularity as a second-line fertility treatment in PCOS. Several modifications have improved its risk-to-benefit profile for patients with PCOS. These modifications include enabling multiple follicular recruitment through gonadotropin stimulation whilst reducing the risk of ovarian hyperstimulation syndrome by using less potent ovulation-triggering agents such as a GnRH agonist. The elective freeze-all strategy, where collected oocytes are fertilized and frozen for later embryo transfer, which allows the ovaries to recover from gonadotropin hyperstimulation, is also employed to prevent exposure to endogenous human chorionic gonadotropin²⁶⁵.

This approach lends itself to elective single embryo transfer of a thawed cryopreserved embryo during a frozen embryo transfer cycle, substantially lowering the likelihood of multiple pregnancies. However, a small chance (<2%) of monozygotic twins exists as a result of elective single embryo transfer²⁶⁶. Although the freeze-all strategy reduces ovarian hyperstimulation syndrome risk and potentially improves pregnancy rates compared with fresh embryo transfer, it may introduce other risks such as an increased incidence of gestational hypertension or pre-eclampsia and large-for-gestational-age babies²⁶⁷. The hypertensive problems may be avoided if frozen embryo transfer is performed in natural cycles with an intact corpus luteum²⁶⁸. Concerns about ovarian hyperstimulation syndrome risk and the excessive use of fertility drugs have prompted the development of in vitro maturation, where oocytes are retrieved after mild ovarian stimulation and matured in vitro before undergoing standard IVF procedures^{269,270}.

Ovarian drilling. The original PCOS treatment by Stein and Leventhal, involving ovarian wedge resection (that is, removal of a wedge-shaped portion in the ovary), improved hyperandrogenism and restored ovulation, enabling natural pregnancy²⁷¹. Modified methods like ovarian diathermy or drilling provide sustained reproductive benefits²⁷², with potentially lower multiple pregnancy rates than low-dose gonadotropin therapy²⁷³. Proponents have developed outpatient ablative procedures to lower costs and increase availability²⁷⁴. The invasiveness of this procedure and the risk of diminishing the ovarian reserve if the operator is inexperienced warrant caution.

Psychological disorders

First-line treatments for PCOS, such as COCPs or lifestyle management, show some improvement or no difference in depressive or anxiety symptoms, suggesting that traditional approaches, including psychological therapy and anti-depressants or anti-anxiety medications, should be used when indicated^{275,276}. As depression, anxiety and eating disorders impact the ability of patients to effectively engage in lifestyle management, screening for these disorders to individualize lifestyle recommendations is needed²⁴. Moreover, the use of cognitive behavioural

therapy along with lifestyle management may be an important strategy to address weight management in those with PCOS and high depressive scores^{276,277}. A multidisciplinary holistic approach that provides education on long-term risks, lifestyle interventions, screening and management of PCOS features is an essential component of improving outcomes, including psychological well-being and HRQoL in patients²⁷⁸.

Quality of life

HRQoL scores related to hirsutism, menstruation and fertility are lower in women with PCOS than in the general population^{279,280}. Identification of areas of key concern that affect quality of life in this multifaceted disorder is critical to prioritizing treatment strategies. However, current surveys do not capture all aspects of PCOS that impact quality of life, and data for adolescents with PCOS is limited. Disease-specific HRQoL questionnaires include questions specific to PCOS, such as menstrual dysfunction, obesity-related matters and how symptoms of hyperandrogenism affect quality of life, and include the validated PCOS health-related QoL questionnaire (PCOSQ)^{281,282} or the PCOS Quality of Life Scale (PCOSQOL). Studies have shown that PCOSQOL measures the impact of PCOS on feminine identity and the negative impact of PCOS on family and friends and on the feeling of pressure to have children in more detail than earlier questionnaires and is sensitive in both clinical and research settings in adult women²⁸³. In addition, the Adolescents' PCOS health-related quality of life Questionnaire (APQ-20) was developed for adolescents and has been shown to be easy to use, valid and reliable²⁸⁴.

PCOS-specific quality of life questionnaires capture the impact of PCOS-specific symptoms on HRQoL, and social, cultural and ethnic differences are observed in the responses²⁷⁹. For example, excess body weight is reported as a common concern in most patient surveys and has a strong association with lower quality of life scores whereas infertility, hirsutism and menstrual symptoms impact quality of life in some studies²⁷⁹. Moreover, surveys in different world regions show that, in some cultures, fertility is a major stigma and fertility-related quality of life scores are more affected; the same holds for hirsutism.

Outlook

Although commendable progress has been made in deciphering the intricacies of PCOS in the past decades, considerable knowledge gaps remain. Our understanding of this complex syndrome is still incomplete and we identify knowledge gaps and urgent research questions (Box 3). Of note, the questions highlighted represent only a fraction of the multifaceted areas that require further investigation. Overall, long-term health, quality of life and ability to work should be at the core of future studies.

Management

Metabolic dysfunction. Currently, data on the impact of specific lifestyle interventions that are effective in the management of increased weight and metabolic dysfunction in PCOS is limited. Preventive therapies to control weight gain in adolescence and peripubertal and periconception interventions are needed. As there is now a growing list of effective anti-obesity agents, the question of how these might be better utilized in weight management and investigation of the long-term metabolic impact of these agents are needed.

Additionally, an improved understanding of the cardiovascular outcomes associated with PCOS and the impact that metabolic dysfunction and its management may have on these outcomes is needed. An understanding of how metabolic dysfunction impacts reproductive function and vice versa is also needed.

Reproductive dysfunction. The genesis of PCOM is unknown and the role of hyperandrogenism in the accumulation of small antral follicles is unclear. Deciphering the intricate relationship between PCOM in the absence of hyperandrogenism and its impact on health and disease is important. Moreover, challenging conventional norms by assessing hyperandrogenaemia in cases where serum levels appear normative is needed.

The accumulation of small antral follicles leads to excessive granulosa cell production of AMH, which in turn increases GnRH and luteinizing hormone secretion, leading to ovarian hyperandrogenaemia and contributing to the development of PCOS; this mechanism needs further investigation. Whether PCOS is of neuroendocrine and/or ovarian origin is still debated, and the respective contribution of each component to the manifestation of the syndrome needs further delineation. The efficacy of neuroendocrine manipulations, including interventions such as kisspeptin, in the treatment of anovulation associated with PCOS needs to be investigated.

Hyperandrogenism. Conducting extensive population-based studies is imperative to comprehensively capture the spectrum of side effects and risks associated with hyperandrogenism treatment in individuals with PCOS. To ascertain the optimal COCP preparation, particularly in adolescents, large-scale comparative studies are essential, focusing on

progestin types and dosages, whilst assessing efficacy on acne, hair loss and psychological outcomes. A thorough investigation into adverse events linked to COCP usage, encompassing aspects such as weight gain, metabolic effects, increased insulin resistance and psychological impact, is imperative. Exploration of the relatively uncharted territory of progestin-only preparations, including intrauterine systems, implants and pills, is warranted to enhance our understanding of their role in treating hyperandrogenism. The potential of anti-obesity drugs in addressing both biochemical and clinical hyperandrogenism, either in isolation or in conjunction with COCPs, necessitates scrutiny through adequately powered randomized trials.

Psychological disorders. Current research strongly supports the integration of psychological well-being in the overall management of PCOS across the lifespan. Routine assessment of mental health conditions in all adolescents and adults evaluated for PCOS is recommended. Further research in understanding the biological, genetic and environmental determinants that impact emotional well-being is urgently needed to better formulate individualized treatment strategies in this high-risk population. Incorporation of strategies to routinely screen patients, incorporate educational materials and access resources might require a change in our current models of care delivery for this chronic condition.

Box 3

Important research questions in PCOS

Management

- Define the best practice lifestyle management to improve reproductive and cardiometabolic risk in polycystic ovary syndrome (PCOS).
- Investigate the effect and mechanisms of new anti-obesity and diabetes drugs on reproductive and cardiometabolic features in PCOS.
- Define the impact of peripubertal weight gain on long-term PCOS risk and interventions that reduce risks.
- Describe interventions for sons of mothers with PCOS.

Reproductive dysfunction

- Origins of polycystic ovary morphology — is hyperandrogenism essential?
- Elucidate the relationship of polycystic ovary morphology in the absence of hyperandrogenism to health and disease.
- Assessment of hyperandrogenaemia when serum levels are normal.
- Define age-specific anti-Müllerian hormone levels as a diagnostic criterion for PCOS.
- Evaluate whether neuroendocrine manipulation, including kisspeptin, is useful in the treatment of anovulation.

Metabolic dysfunction

- Delineate the role of metabolic dysfunction in cardiovascular risk in PCOS.
- Define reliable tools for assessing cardiovascular risk in women with PCOS.

- Understand molecular mechanisms of insulin resistance, oxidative stress and mitochondrial dysfunction.
- Define the role of the immune system in target tissues, including reproductive and metabolic tissues.

Inheritance and mechanisms

- Identify the effector gene at each locus and define how the altered expression or function of each gene contributes to PCOS pathophysiology.
- Define the genetic architecture of PCOS in African or Hispanic populations.
- Explore and define subtypes of PCOS.
- Delineate neuroendocrine versus ovarian origins of PCOS.
- Define key triggers (for example, androgens and anti-Müllerian hormone) and molecular pathways that drive PCOS-specific reproductive, cardiometabolic and psychiatric dysfunction.
- Define how PCOS affects specific cell types in key target tissues such as ovary, endometrium, adipose tissue, skeletal muscle and liver.
- Identify biomarkers to allow early diagnosis and development of novel drug targets as well as predictive markers.
- Define targeted prenatal interventions to influence fetal programming.
- Define the male phenotype of PCOS for the assessment of male relatives of women with PCOS.

Mechanisms

Although GWAS has unveiled >20 genetic variants linked to PCOS, crucial gaps persist in understanding the effector genes at these loci and in deciphering how the altered expression or function of each gene contributes to PCOS pathophysiology. Although large-scale GWAS in PCOS are prevalent in European and Chinese cohorts, the genetic architecture in other populations, like African or Hispanic, remains inadequately explored. Investigating potential PCOS subtypes necessitates analysing larger cohorts for statistical robustness during sample subdivision. In the realm of inheritance, exploring the contribution of epigenetic processes in longitudinal mother–child cohorts is crucial for understanding maternal–fetal factors in PCOS transmission. Utilizing PCOS-like mouse models offers insight into key drivers and disease transmission loci.

A deeper understanding of the underlying mechanisms of compensatory hyperinsulinaemia, insulin resistance, T2DM, systemic inflammation, cardiovascular disease and pregnancy complications in PCOS is urgently needed. Employing temporal and spatial single-cell technologies for ovaries, endometrium, adipose tissue, skeletal muscle, liver, pancreas and brain has the potential to unveil complex cellular interactions, identify biomarkers for prediction and targeted treatment, and predict treatment response.

Epidemiology

Large data sets and epidemiological studies are central to understanding the features and natural history of PCOS²⁸⁵. Epidemiological data, including routine clinical data, are also key to evaluating care and treatment responses across health-care settings²⁸⁶. Yet, only few high-quality longitudinal data sets in unselected populations with intergenerational, early and later life event data are available. Future and existing data sets should identify PCOS status, including self-report, PCOS symptoms and, ideally, androgen and AMH measurement²⁸. Data harmonization and linkage are also vital to address current gaps and explore heterogeneity and ethnicity²⁸⁷. Data harmonization, core outcome set integration, secure data sharing, federated data analyses, and the use of artificial or synthetic data, alongside advanced analytics to harness the power of data in PCOS are warranted.

Gaps identified by updated guidelines

Research priorities identified in the 2023 International Evidence-based Guideline for the Assessment and Management of PCOS¹ focus on defining diagnostic cut-offs, developing prevention strategies, addressing psychological aspects associated with PCOS, and devising optimal models of care and novel medical and surgical treatments for the management of PCOS in individuals with excess body weight. The international research priorities are based on surveys of thousands of health professionals and women with PCOS^{18,288} and the process involved analysis of 52 systematic reviews²⁴ and consensus processes via the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework²⁸⁹. Clinical expertise, consumer preference and evidence gaps were captured, and the final consensus was obtained across guideline experts from 6 continents, representing 39 engaged organizations¹.

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The authors declare no competing interests.

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¹Department of Physiology and Pharmacology, Karolinska Institute, Stockholm, Sweden. ²Monash Centre for Health Research and Implementation, Monash Health and Monash University, Melbourne, Victoria, Australia. ³Robinson Research Institute, Adelaide Medical School, Adelaide, South Australia, Australia. ⁴Department of Obstetrics and Gynecology, Penn State College of Medicine, Hershey, PA, USA. ⁵Department of Public Health Science, Penn State College of Medicine, Hershey, PA, USA. ⁶Division of Endocrinology, Diabetes and Metabolism, Department of Medicine, Cedars-Sinai Medical Center, Los Angeles, CA, USA. ⁷Department of Obstetrics and Gynecology, University of Pennsylvania, Philadelphia, PA, USA. ⁸Division of Reproductive Endocrinology & Infertility, Department of Obstetrics and Gynecology, Erasmus MC, Rotterdam, Netherlands. ⁹Department of Obstetrics and Gynecology, University of Rochester School of Medicine and Dentistry, Rochester, NY, USA. ¹⁰Department of Obstetrics and Gynecology, Research Unit of Clinical Medicine, Medical Research Center, Oulu University Hospital, University of Oulu, Oulu, Finland.